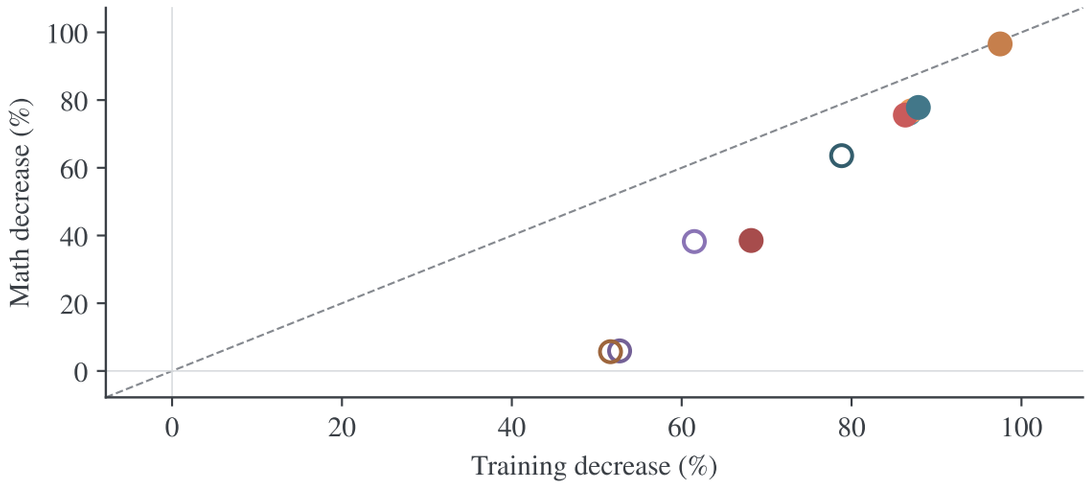
Qwen3-1.7B / non-thinking | Training pool vs. held-out

Relative RKL decrease (%)

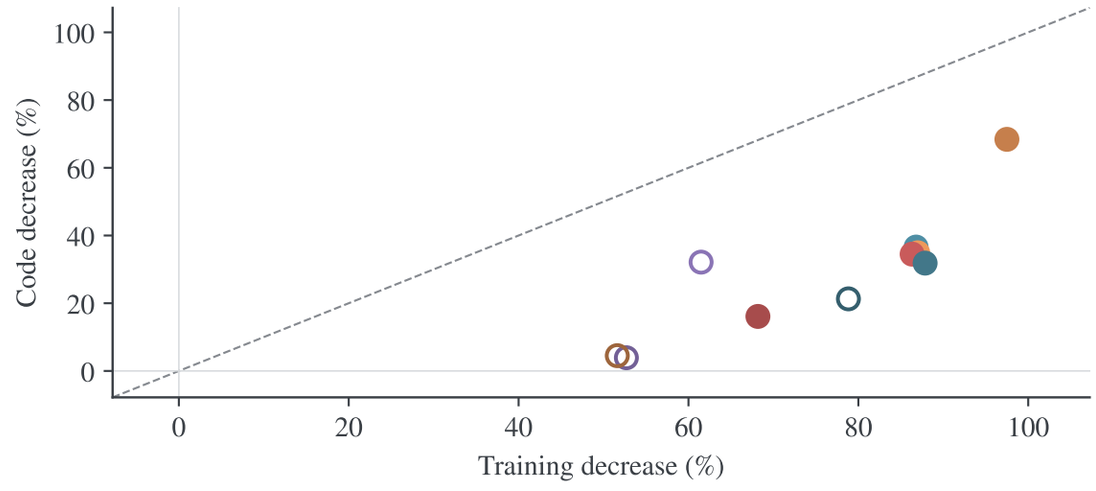
(a) Training vs. Math

n=10; r=0.99; rho=1.00



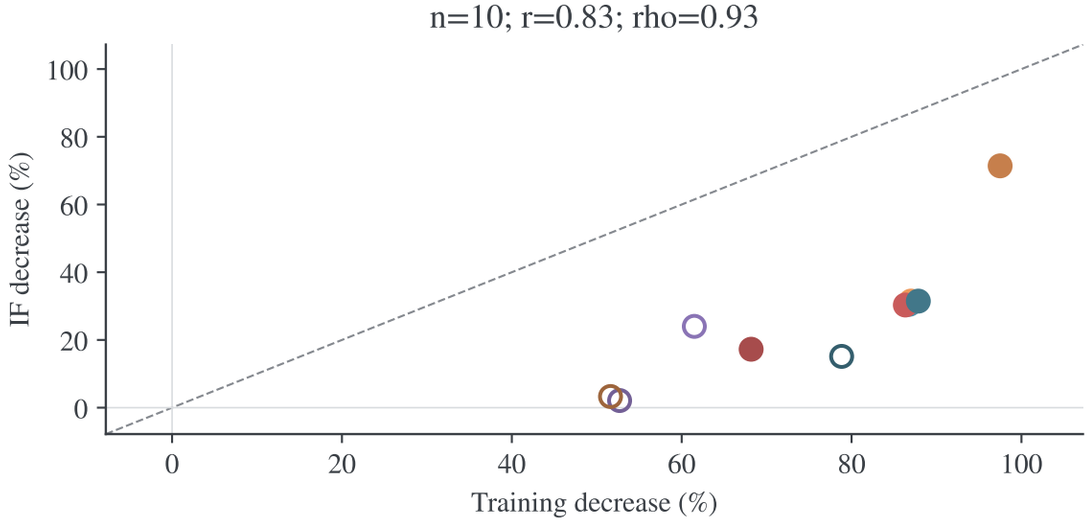
(b) Training vs. Code

n=10; r=0.84; rho=0.79



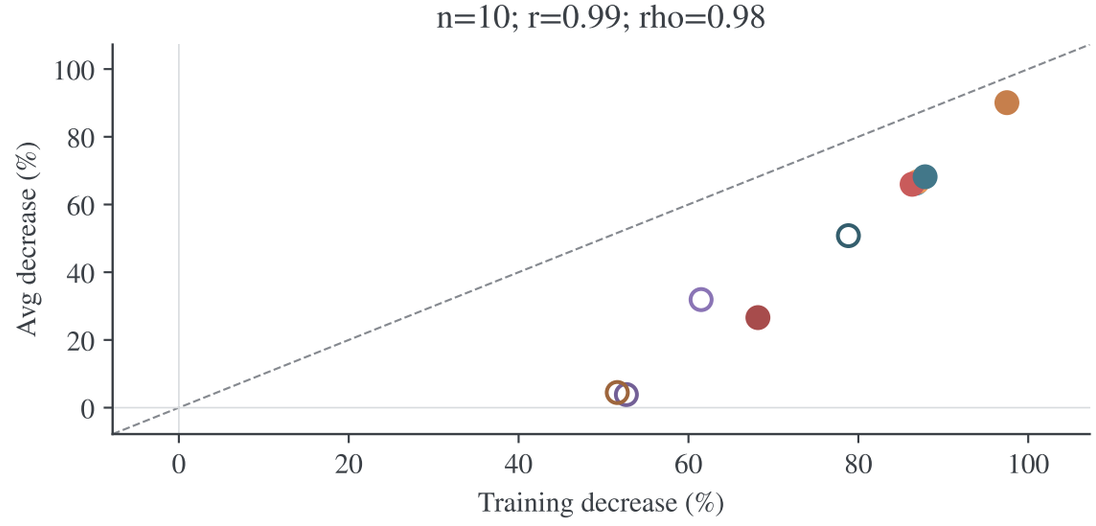
(c) Training vs. IF

n=10; r=0.83; rho=0.93



(d) Training vs. Avg

n=10; r=0.99; rho=0.98



- Co-rewarding-I-Qwen3-1.7B-Base-MATH [320]
- Co-rewarding-II-Qwen3-1.7B-Base-MATH [320]
- GT-Qwen3-1.7B-Base-MATH [320]
- Qwen3-0.6B [320]
- Qwen3-1.7B-Base-OPD / Thinking-Space [320]
- Qwen3-1.7B-Math / Samsung SFT [320]
- Qwen3-1.7B-SFT / Thinking-Space [320]
- Qwen3-4B [320]
- Qwen3-4B-Base-GRPO / Thinking-Space [320]
- Qwen3-8B [320]
- Equal decrease

One point = one run endpoint; triangle: 64 updates; circle: 320; filled: same size; open: cross size.

Fixed-H RKL; dashed: equality; correlations pool all displayed endpoints (mixed budgets/sizes).